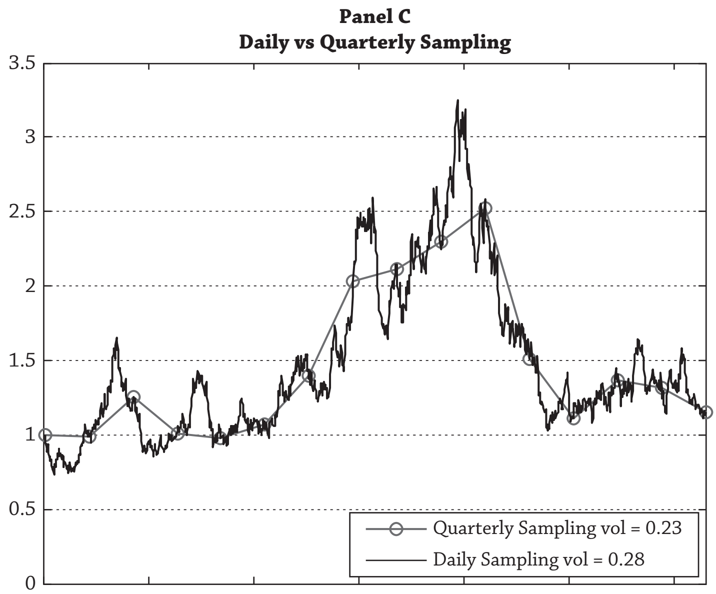

Figure 13.2 (continued)

of each quarter. I produced the graphs in Figure 13.2 by simulation and deliberately chose one sample path where the prices have gone up and then down to mirror what happened to equities during the 2000s' Lost Decade. The prices in Panel A appear to be drawn from a series that does not seem excessively volatile; the standard deviation of quarterly returns computed using the prices in Panel A is 0.23.

The true daily returns are plotted in Panel B of Figure 13.2 These are much more volatile than the ones in Panel B. Prices go below 0.7 and above 3.0 in Panel B with daily sampling, whereas the range of returns in Panel A is between 1.0 and 2.5 with quarterly sampling. The volatility of quarterly returns, computed from (overlapping) daily data in Panel B is 0.28, which is higher than the volatility of quarterly-sampled returns of 0.23 in Panel A.

For a full comparison, Panel C plots both the quarterly and daily sampled returns and just overlays Panels A and B in one picture. Infrequent sampling has caused the volatility estimate using the quarterly sampled returns to be too low. The same effect also happens with betas and correlations—risk estimates are biased downward by infrequent sampling.[11]

[11] See Geltner (1993) and Graff and Young (1996) for infrequent observation bias on the effect of betas and correlations, respectively. Geltner estimates that betas are understated by a factor of 0.5 for real estate returns. This is not a "small sample" problem, which goes away when our sample becomes very large; it is a "population" problem as the next section explains.